DS6410 MACHINE LEARNING II — LECTURE 16

Directed Graphical Models

Causality, Inference, and Applications

Bayesian Networks · Do-Calculus · Mediation Analysis · Mendelian Randomization

From Undirected to Directed: Why Direction Changes Everything

UNDIRECTED MODELS (MRFS)

- **Symmetric Associations:** Edges denote mutual (partial) correlation.
- **Factorization:** Factorizes over cliques; requires a global partition function Z.
- **Limitation:** Cannot represent asymmetric (causal) directions.

Use Case

Good for spatial/image segmentation or co-expression networks where direction doesn't matter.

DIRECTED MODELS (BAYESIAN NETWORKS)

- **Asymmetric Relationships:** Directed edges encode conditional (often causal) dependencies.
- **Factorization:** Factorizes locally over parent–child relations; no global Z needed.
- **Generative Process:** Describes stepwise data generation (ancestral sampling).

Use Case

Useful for causal inference, expert systems, and sequential/hierarchical data models.

General Factorization

The joint distribution of K variables factorizes as:

$$p(x_1, \ldots, x_K) = \prod_{k=1}^{K} p(x_k \mid \mathrm{pa}_k)$$

Chain Rule of Probability

$$p(x_1, \ldots, x_K) = \prod_{k=1}^{K} p(x_k \mid x_1, \ldots, x_{k-1})$$

Bayesian Networks: DAGs and the Factorization Theorem

A **Bayesian Network** is a Directed Acyclic Graph (DAG) where nodes represent random variables and directed edges encode direct probabilistic dependencies.

- **Acyclic Property:** The graph must not contain any directed cycles. This ensures the factorization represents a valid probability distribution.
- **Sparsity:** By dropping edges from the fully connected chain rule, the DAG encodes conditional independencies, reducing the number of parameters required.

INTERPRETATION OF EDGES

An edge from node A to node B indicates that A is a parent of B. The absence of an edge implies conditional independence: each variable is conditionally independent of its non-descendants given its parents.

Factorization in Practice: A Concrete DAG Example

A 5-NODE NETWORK

Consider a simple DAG with 5 binary variables:

A → C ← B ↓ ↓ D E

Joint Distribution Factorization:

$$P(A, B, C, D, E) = P(A)\,P(B)\,P(C \mid A, B)\,P(D \mid C)\,P(E \mid C)$$

PARAMETER REDUCTION

Assume all variables are binary (0 or 1). How many independent parameters are needed to specify the joint distribution?

Component	Parameters Needed
P(A)	1
P(B)	1
P(C | A, B)	4 (one for each state of A, B)
P(D | C)	2 (one for each state of C)
P(E | C)	2 (one for each state of C)
Total (Factorized)	**10 parameters**

The Curse of Dimensionality

A full joint for 5 binary variables needs 2^5 - 1 = 31 parameters. The DAG reduces this to 10, a dramatic savings for larger networks.

Three Canonical Structures: The Building Blocks of DAGs

Every path in a Bayesian Network is composed of these three fundamental motifs.

HEAD-TO-TAIL (CHAIN)

A → C → B

Node C acts as a **mediator** between A and B. A influences B only through C.

$$A \not\perp\!\!\!\perp B$$

$$A \perp\!\!\!\perp B \mid C$$

> **Rule:** Conditioning on the middle node C **BLOCKS** the path, making A and B independent.

TAIL-TO-TAIL (FORK)

A ← C → B

Node C acts as a **common cause** (confounder) of both A and B.

$$A \not\perp\!\!\!\perp B$$

$$A \perp\!\!\!\perp B \mid C$$

> **Rule:** Conditioning on the common cause C **BLOCKS** the path, making A and B independent.

HEAD-TO-HEAD (COLLIDER)

A → C ← B

Node C is a **common effect** (collider) of A and B. This forms a v-structure.

$$A \perp\!\!\!\perp B$$

$$A \not\perp\!\!\!\perp B \mid C$$

> **Rule:** Conditioning on the collider C (or its descendants) **OPENS** the path, making A and B dependent.

The V-Structure

$$a \rightarrow c \leftarrow b$$

Node c is a "collider" because two arrows collide at it.

Marginal Independence

Without observing c, the path is blocked:

$$a \perp\!\!\!\perp b$$

Conditional Dependence

Observing c (or any descendant of c) opens the path:

$$a \not\perp\!\!\!\perp b \mid c$$

Explaining Away: The Collider Opens a Path When Conditioned On

The **collider** (or head-to-head node) exhibits a counter-intuitive property unique to directed graphs. Two independent causes become dependent when we observe their common effect.

- **The Mechanism:** Knowing the effect and finding Cause A did *not* happen makes Cause B more likely to "explain away" the effect.
- **Selection Bias:** Conditioning on a collider often happens during data collection, creating spurious correlations.

> **REAL-WORLD EXAMPLE: BERKSON'S PARADOX**
>
> Suppose Talent (a) and Attractiveness (b) are independent but both increase the chance of becoming an Actor (c). If we only look at actors, talent and attractiveness will appear negatively correlated: if an actor is not talented, they tend to be more attractive.

D-Separation Definition

Sets of nodes A and B are d-separated by Z if every path from a node in A to a node in B is blocked by Z.

$$A \perp\!\!\!\perp B \mid Z$$

Soundness & Completeness

Soundness: If A and B are d-separated by Z, they are conditionally independent in all distributions that factorize over the DAG.

Completeness: If they are not d-separated, there exists a distribution where they are dependent.

D-Separation: Reading Conditional Independence from the Graph

Check whether all paths between two variables are **blocked** by the conditioning set Z. A path is blocked if it contains a node m such that:

- **Rule 1 (Chain or Fork):** Path has $i \to m \to j$ or $i \leftarrow m \to j$, and the middle node m is in Z.
- **Rule 2 (Collider):** Path has $i \to m \leftarrow j$, and neither m nor its descendants are in Z.

> **THE KEY INSIGHT**
>
> D-separation is a graphical criterion that lists the model's conditional independencies directly from the graph.

Factor Graphs: A Unified Language for Graphical Models

BIPARTITE STRUCTURE

A **Factor Graph** makes factorization explicit via a bipartite graph with two node types.

- **Variable Nodes (Circles):** Represent variables x_i.
- **Factor Nodes (Squares):** Represent local factors f_s.

Edges connect variable nodes only to factor nodes when the variable is an argument of the factor.

> **Why Factor Graphs?**
> They expose the exact factorization and are the standard representation for inference.

THE UNIFYING FRAMEWORK

Joint distribution = product of factors:

$$p(\mathbf{x}) = \prod_s f_s(\mathbf{x}_s)$$

where $\mathbf{x}_s$ is the subset of variables for factor f_s.

- **Directed Models:** Each factor corresponds to p(x_i | pa_i).
- **Undirected Models (MRFs):** Each factor is a clique potential $\psi_C(\mathbf{x}_C)$.

Inference: The Variable Elimination Algorithm

EXACT INFERENCE

The goal of inference is to compute a marginal distribution, e.g., P(X), by summing out other variables from the joint distribution.

Naive Approach:

$$P(X) = \sum_{Y,Z} P(X, Y, Z)$$

Requires forming the full joint table (exponential).

Variable Elimination:

$$P(X) = \sum_{Y} P(Y) \sum_{Z} P(X \mid Y, Z) P(Z)$$

Push sums inward to work on local factors only.

COMPUTATIONAL COMPLEXITY

- **Intermediate Factors:** Eliminating variables creates new factors.
- **Cost:** Exponential in the size of the **largest intermediate factor**.
- **Treewidth:** Largest factor size (minus 1) under the best ordering.

The Role of Elimination Ordering

The elimination order greatly affects intermediate factor size. Finding the optimal ordering is **NP-hard**; heuristics like min-fill or min-degree are used.

Structure Learning: Discovering DAGs from Observational Data

THE PROBLEM

So far, we have assumed the DAG structure is **known**. In practice, we often need to **learn the graph from data** using conditional independence tests — no interventions required.

TWO MAIN PARADIGMS

- **Constraint-Based:** Use statistical tests of conditional independence to remove and orient edges (leverages d-separation directly).
- **Score-Based:** Define a penalized score (e.g., BIC) and search the space of DAGs to maximize it.

KEY ALGORITHMS

- **PC Algorithm:** Constraint-based; assumes no unmeasured confounders. Outputs a CPDAG.
- **FCI:** Robust to unmeasured confounders and selection bias. Outputs a PAG.
- **GES:** Score-based search over Markov equivalence classes; asymptotically consistent.

TRANSITION: FROM STRUCTURE TO CAUSATION

These algorithms recover the **skeleton and orientation** of a DAG from observational data alone. But answering **causal questions** — "What happens if I intervene?" — requires a new language: **do-calculus**.

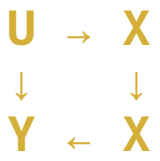
From Association to Causation: The do-Operator

OBSERVATIONAL: "SEEING"

$$P(Y \mid X = x)$$

Meaning: "What is the probability of Y given that we **observe** $X = x$?"

This is standard conditional probability. We filter the existing population to only those individuals where $X = x$ naturally occurred.



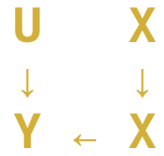
(Original Graph G)

INTERVENTIONAL: "DOING"

$$P(Y \mid do(X = x))$$

Meaning: "What is the probability of Y if we **force** X to take the value x?"

This represents a physical intervention (like a randomized experiment). We override the natural causes of X.



(Mutilated Graph $G_{\overline{X}}$)

The Fundamental Problem of Causal Inference

If there are unmeasured confounders (U), then **conditioning is not the same as intervening.**

The Three Rules of Do-Calculus

A formal algebraic system for transforming interventional probabilities.

RULE 1: OBSERVATIONS

Insertion/Deletion of Observations

$$P(y \mid do(x), z, w) = P(y \mid do(x), w)$$

We can ignore observation Z if it provides no new information about Y given X and W in the mutilated graph.

Condition:
$Y \perp\!\!\!\perp Z \mid X, W$ in $G_{\overline{X}}$

RULE 2: EXCHANGE

Action/Observation Exchange

$$P(y \mid do(x), do(z), w) = P(y \mid do(x), z, w)$$

We can replace an intervention $do(Z)$ with an observation Z if the intervention does not block any backdoor paths.

Condition:
$Y \perp\!\!\!\perp Z \mid X, W$ in $G_{\overline{X}\underline{Z}}$

RULE 3: ACTIONS

Insertion/Deletion of Actions

$$P(y \mid do(x), do(z), w) = P(y \mid do(x), w)$$

We can completely remove an intervention $do(Z)$ if there is no causal path from Z to Y.

Condition:
$Y \perp\!\!\!\perp Z \mid X, W$ in $G_{\overline{XZ(W)}}$

THE COMPLETENESS THEOREM (SHPITSER & PEARL, 2006)

Do-calculus is **complete**. If a causal effect $P(Y \mid do(X))$ can be identified from observational data and the DAG structure, it can be derived using a finite sequence of these three rules. If do-calculus cannot identify the effect, no method can.

Backdoor Criterion: Blocking Confounding Paths

THE CRITERION

A set of variables **Z** satisfies the **backdoor criterion** relative to (X, Y) if it removes confounding into X.

- No node in **Z** is a descendant of **X**.
- **Z** blocks every path between **X** and **Y** that has an arrow into **X**.

Backdoor Adjustment Formula:

If Z satisfies the criterion, the causal effect is:

```
P(Y | do(X=x)) = sum_z P(Y | X=x, Z=z) P(Z=z)
```

GRAPHICAL IDENTIFICATION

The backdoor criterion gives a graphical rule for which variables to condition on to remove confounding.

- **Backdoor Paths:** Paths like X ← … → Y that induce spurious correlation.
- **Blocking:** Use d-separation to ensure Z blocks these paths.
- **No Descendants:** Do not condition on descendants of X (mediators).

Why This Matters

The adjustment uses only **observational** terms P(Y|X,Z) and P(Z). Thus we can estimate the interventional distribution P(Y | do(X)) from observed data when a valid Z is measured.

Frontdoor Criterion: Identifying Effects Through Mediators

When unmeasured confounders (U) exist, the backdoor criterion fails because we cannot condition on U. However, a **mediator** M that satisfies the frontdoor conditions lets us identify the causal effect.

- **Step 1:** Identify the effect of X on M: $P(M \mid do(X)) = P(M \mid X)$.
- **Step 2:** Identify the effect of M on Y; block backdoor paths by conditioning on X.
- **Step 3:** Combine these using the frontdoor adjustment.

FRONTDOOR ADJUSTMENT FORMULA

$$P(Y \mid do(X = x)) = \sum_m P(M = m \mid X = x) \sum_{x'} P(Y \mid X = x', M = m) P(X = x')$$

Like the backdoor formula, this uses observational data, bypassing the unmeasured confounder U.

The Frontdoor DAG

```
      U
    ↙   ↘
X → M → Y
```

U is an unmeasured confounder.

The Three Conditions

A set of variables M satisfies the frontdoor criterion relative to (X, Y) if:

1. M intercepts all directed paths from X to Y.
2. There is no unblocked backdoor path from X to M.
3. All backdoor paths from M to Y are blocked by X.

Mediation Analysis: Decomposing Total Effects

THE MEDIATION DAG

```
      M
    ↗   ↘
X     →     Y
```

Mediation analysis studies **how** an exposure **X** affects an outcome **Y** by splitting the total effect into direct and indirect parts.

- **Direct Effect:** Path X → Y (not via M).
- **Indirect Effect:** Path X → M → Y.

$$\text{Total Effect} = \text{Direct Effect} + \text{Indirect Effect}$$

CLASSICAL BARON-KENNY FRAMEWORK

The traditional approach uses three regressions to identify paths.

1. $M = \alpha + aX + \varepsilon 1$
2. $Y = \beta + bM + c'X + \varepsilon 2$
3. $Y = \tau + cX + \varepsilon 3$

- **Indirect Effect:** $a \times b$.
- **Direct Effect:** c'.
- **Total Effect:** $c = c' + a \times b$.

Limitations

Assumes linearity and **no interaction** between X and M. If interaction exists, $a \times b$ may misrepresent the indirect effect.

Causal Mediation: Natural Direct and Indirect Effects

PEARL'S FRAMEWORK

Using potential outcomes notation $Y(x, m)$, we can define effects counterfactually, allowing for non-linearities and interactions.

Natural Direct Effect (NDE):

Effect of changing X from x_0 to x, keeping M at the value it *would have taken* under x_0.

$$\text{NDE} = E[Y(x, M(x_0)) - Y(x_0, M(x_0))]$$

Natural Indirect Effect (NIE):

Effect of changing M from $M(x_0)$ to $M(x)$, keeping X fixed at x.

$$\text{NIE} = E[Y(x, M(x)) - Y(x, M(x_0))]$$

IDENTIFICATION ASSUMPTIONS

To identify NDE and NIE from observational data, four strong assumptions are required:

- No unmeasured confounding of the $X \to Y$ relationship.
- No unmeasured confounding of the $M \to Y$ relationship.
- No unmeasured confounding of the $X \to M$ relationship.
- **No treatment-induced confounding:** There is no $M \to Y$ confounder that is itself affected by X.

Why Classical Mediation Fails

The Baron-Kenny method (Total Effect = $c' + a \times b$) assumes linear models without interactions. If the effect of the mediator M on Y depends on the level of X (an interaction), the product $a \times b$ no longer represents a valid causal indirect

Mendelian Randomization: Genetics as a Natural Experiment

THE MOTIVATION

Estimating causal effects in observational data is notoriously difficult due to **unmeasured confounding** (e.g., lifestyle, socioeconomic status).

MENDEL'S LAWS

- **Random Assortment:** Alleles (genetic variants) are randomly assigned from parents to offspring at conception.
- **Independence:** This assignment is generally independent of environmental and lifestyle confounders that develop later in life.

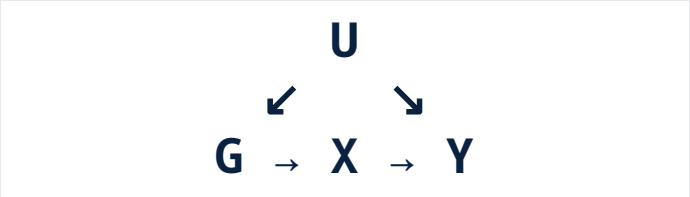
INSTRUMENTAL VARIABLES (IVS)

MR is an application of **Instrumental Variable (IV)** analysis from econometrics.

- **The Instrument:** We use Single Nucleotide Polymorphisms (SNPs) that are robustly associated with the exposure of interest (e.g., a SNP known to increase LDL cholesterol).
- **The Logic:** If the exposure causally affects the outcome, then the genetic variant (which acts as a proxy for the exposure) should also be associated with the outcome.

MR: DAG Structure, IV Assumptions, and Estimation

The MR DAG



G: Genotype (IV), X: Exposure, Y: Outcome, U: Confounder.

Three Core IV Assumptions

1. **Relevance:** The genetic variant G is robustly associated with the exposure X ($G \not\perp\!\!\!\perp X$).
2. **Independence (Exogeneity):** G is independent of all confounders U ($G \perp\!\!\!\perp U$).
3. **Exclusion Restriction:** G affects the outcome Y **only** through the exposure X (no direct path $G \to Y$).

ESTIMATION METHODS

If the three IV assumptions hold, we can estimate the causal effect of X on Y.

- **Wald Estimator (Single IV):** The causal effect β is the ratio of the gene-outcome association to the gene-exposure association: $\beta_{IV} = \frac{\mathrm{Cov}(Y,G)}{\mathrm{Cov}(X,G)}$
- **Two-Stage Least Squares (2SLS):** Used when individual-level data is available. Stage 1: Regress X on G to get predicted $\hat{X}$. Stage 2: Regress Y on $\hat{X}$.

TWO-SAMPLE MR AND IVW

Modern MR often uses summary statistics from two independent GWAS (one for $G \to X$, one for $G \to Y$). The **Inverse Variance Weighted (IVW)** method combines multiple independent SNPs by taking a meta-analysis of their individual Wald ratios, weighted by their precision.

MR Sensitivity Analysis and Violations

THE MAIN THREAT: PLEIOTROPY

Pleiotropy occurs when a single genetic variant influences multiple traits.

- **Vertical Pleiotropy:** SNP affects outcome via the exposure (G → X → Y).
- **Horizontal Pleiotropy:** SNP affects outcome via an independent pathway (violates exclusion).

WEAK INSTRUMENT BIAS

Occurs when selected SNPs weakly predict the exposure.

- One-sample MR: bias toward the observational (confounded) estimate.
- Two-sample MR: bias toward the null.

ROBUST ESTIMATION METHODS

Sensitivity analyses relax strict IV assumptions to guard against horizontal pleiotropy.

- **MR-Egger:** Allows pleiotropy; intercept detects directional effects.
- **Weighted Median:** Consistent if ≤50% of weight comes from invalid instruments.
- **MR-PRESSO:** Detects and removes outlier SNPs with horizontal pleiotropy.

> **Triangulation**
>
> No single method is perfect. Use multiple methods; concordant results increase confidence.

The Ladder of Causation

3. Counterfactuals

Imagining, Retrospection

$$P(y_x \mid x', y')$$

"What if I had acted differently?"

2. Intervention

Doing, Intervening

$$P(y \mid do(x))$$

"What if I do X?"

1. Association

Seeing, Observing

$$P(y \mid x)$$

"What if I see X?"

Summary: Directed Graphical Models as a Causal Language

We have seen how Directed Acyclic Graphs (DAGs) provide a rigorous mathematical language for moving from mere association to true causal inference.

- **DAG Factorization & d-separation:** The foundational rules that link the graphical structure to conditional independencies in the data.
- **The do-Calculus:** A complete algebraic system for transforming interventional probabilities into observational ones, allowing us to identify causal effects from observational data.
- **Backdoor & Frontdoor Criteria:** Practical graphical rules for identifying valid adjustment sets to block confounding paths or leverage mediators.
- **Mediation Analysis:** Decomposing total effects into direct and indirect pathways, formalized via Natural Direct and Indirect Effects.
- **Mendelian Randomization:** Using genetics as instrumental variables to perform "Nature's RCTs" and overcome unmeasured confounding.